Akash Dwivedi

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SUMMARY

Software Engineer and final-year Computer Science student (Class of 2027) with a strong foundation in **Data Structures & Algorithms**, **System Design**, and **object-oriented programming**. Experienced in building **scalable** web applications and backend architectures using **Java**, Spring Boot, Next.js, Node.js, and **PostgreSQL**. Hands-on with **RESTful API design**, database optimization, **concurrency** control, JWT authentication, and real-time AI integrations. LeetCode Knight (rating 1873) with a strong track record of solving complex, ambiguous technical problems.

EXPERIENCE

Software Developer Intern — ChatFin.ai (Agentic AI Platform for Finance) *Sep 2026 – Present*

Internship | TypeScript, Node.js, Express, MySQL, MCP, LLM Tool Calling, Slack Bolt, REST APIs

- Worked on **lazy auth** for agent tool calls: a plugin call that needs a user login suspends in place, sends a login link, and resumes once the user signs in, so users log in only when a plugin first needs it.
- Helped build an **event-driven Slack architecture**: channel and DM messages, with files, are raised on an **event bus** where subscribed workflows and apps respond, falling back to the AI agent; upgraded the Slack Bolt/Express layer.
- Built **plugins** that AI agents call as tools and MCP clients such as Claude discover and run through our **MCP server**, including a **NetSuite** plugin that posts vendor bills from chat by mapping a JSON-schema tool to SuiteTalk REST.
- Re-architected the agents' email layer into a **multi-provider mail service** (Gmail, Outlook, SMTP, IMAP) with named mailboxes the LLM picks as **tool arguments**, letting one deployment serve many clients; built the IMAP reader and mailbox routing.
- Built features and fixed bugs in live **client finance workflows** (AR dunning on SAP Business One, cash application), including a **production MySQL failure** traced to array bind parameters and fixed in the shared DB layer for every query.
- Hardened the **workflow engine** (inputs survive failed runs; failures logged with instance and raw error) and fixed dependency **vulnerabilities** across the backend, MCP server and UI (Slack Bolt, Express, MCP SDK upgrades).

Freelance Software Developer — Shiksha (Digital Log Management Tool) *Nov – Dec 2025*

Freelance Client Project · Remote | Next.js, Node.js, PostgreSQL, Redis, JWT

- Delivered a freelance transaction & inventory platform with 12+ modular **REST API** endpoints (sales, purchases, ledger, reporting); automated 3 reporting frequencies with running-balance **algorithms**, cutting reconciliation effort by 70%.
- Implemented **PostgreSQL**-backed transaction & credit-tracking with Redis caching and JWT authentication; resolved 15+ critical bugs before delivery.

PROJECTS

BaseCase – AI-Powered DSA Learning & Interview Prep Platform | Next.js, Node.js, PostgreSQL, Qdrant, Redis, OpenAI SDK *Jan 2026 – Ongoing*

Link: basecase-xi.vercel.app | GitHub: github.com/aakash-1702/basecase

- Engineered the platform's core **AI Voice Interviewer** with **RAG-based grounding** over Qdrant vector embeddings and a 4-state FSM (idle → listening → processing → speaking) via the Web Speech API, powering 100+ context-aware mock interviews.
- Built a GitHub-based **AI interview mode** that analyzes candidate repositories with LLM-driven code understanding to auto-generate personalized, codebase-grounded questions.
- Integrated **OpenAI SDK**-powered quiz generation with SM-2 spaced repetition across 200+ DSA problems; self-hosted Judge0 for secure code execution, scaling to 1,000+ users.

QueueCTL – CLI-Based Distributed Job Queue System | Node.js, TypeScript, PostgreSQL, Prisma *2026*

- Built a **decentralized, database-driven job queue** eliminating the need for a central broker (Redis/RabbitMQ), using PostgreSQL's FOR UPDATE SKIP LOCKED for lock-free **concurrent** job claiming across workers.
- Architected a **fault-tolerant** worker system using Node's child_process.fork() with exponential-backoff retry **algorithms** and a Dead-Letter Queue for exhausted jobs.
- Designed a cross-platform graceful-shutdown mechanism (DB signaling, IPC, SIGTERM) and validated the **distributed system** with a 5-scenario end-to-end suite covering concurrency and crash recovery.

CERTIFICATIONS

- **Agentic AI Certified Foundations Associate** — Oracle
- **Advanced RAG with Vector Databases and Retrievers** — Coursera
- **Developing Back-End Apps with Node.js and Express** — Coursera
- **SQL & Data Engineering Foundations** — TUF

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C++, SQL
Frontend: React.js, Next.js, HTML5, CSS3
Backend: Spring Boot, Spring MVC, Spring Data JPA, Node.js, Express.js, REST APIs, Apache Kafka
Databases: MySQL, PostgreSQL, MongoDB, Redis, Qdrant
Concepts: OOP, Data Structures & Algorithms, System Design, API Design, Secure Coding, CN, DBMS, OS
Testing & Tools: Git, GitHub, JUnit, Mockito

ACHIEVEMENTS

LeetCode Knight – Maximum Rating 1873 | **CodeChef 3-Star Coder**
Ranked 22nd out of 500+ participants in a college-level Hackathon

EDUCATION

Pranveer Singh Institute of Technology *2023 – 2027*
B.Tech in Computer Science and Engineering — 79.16% *Kanpur, India*